



Breaking a Sweat in VR: What Changes When "Gaming" Becomes Active

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ABSTRACT

- This study examined differences in time estimation accuracy, engagement, and subjective experience between participants engaging in a Virtual Reality (VR) task and those playing a standard video game. By comparing a kinesthetic virtual reality experience with a traditional controller-based video game experience, this research aims to examine how physical interaction impacts the user experience.

BACKGROUND

- Prior studies have suggested that VR alters spatial awareness, presence, and sensory integration. For instance, studies have demonstrated that VR-based simulations can effectively enhance student motivation compared to traditional pedagogical methods (Makransky et al., 2019). However, many of these studies involve stationary activities where the focus is on interaction with the game rather than physical performance. While previous research has focused on the psychological effects of VR gaming, particularly over immersion and motivation, there is a notable gap in the research about the influence of *kinesthetic engagement* (i.e., full-body physical interaction) on persistence in VR games. Studies around sports video games have highlighted motivation and achievement, but few have explored dynamics within VR settings that demand more physical activity.
- This study seeks to examine how the physical aspect of playing a VR tennis-like video game impact various psychological factors. Understanding these influences could have broader implications for the design of engaging VR games and applications.

METHOD

Participants

- The study was performed with $n = 34$ Psychology-Major, Undergraduate students at a small, private university in Fort Worth, Texas.

Group Assignments

- Group 1 engaged in a VR tennis-like game, requiring participants to physically stand and swing their arms in a simulated tennis match. Group 2 played a traditional controller-based tennis game while seated at a desk, where physical interaction was minimal. Both groups played for approximately 13 minutes.

Data Collection

- Participants answered demographic questions, as well as questions about their experience during and after gameplay.

Gameplay Sessions and Question Blocks

- During the gameplay sessions, participants were interrupted at different intervals (4, 8, and 13 minutes) to answer a short series of questions. This process was repeated three times, resulting in a total of three question blocks throughout the task, followed by a longer survey post-gameplay.

Figure 1. In-Game Survey Questions

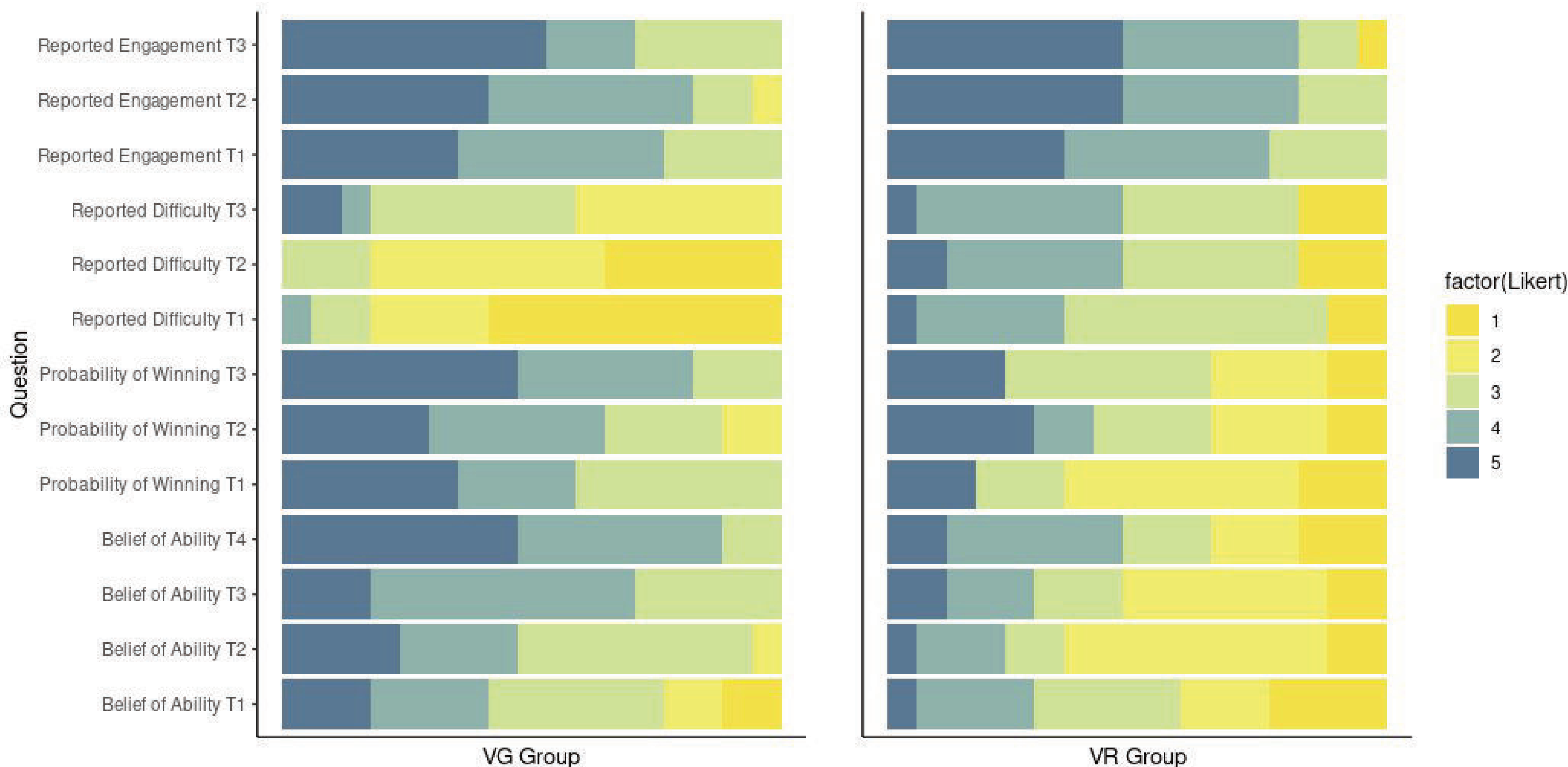


Figure 2. Post-Game Survey Questions



RESULTS

- Differences in Duration Estimation:** The two groups did not show any significant difference in their ability to estimate how much time has passed. See Table 1.
- Qualitative Survey Differences:** Though there are minor differences, we can see **similar patterns** of the self-report surveys between groups:
 - Engagement** goes up with each intake, but most participants reported "high engagement" across report intervals (4 minutes, 8 minutes, 13 minutes) for both groups. This shows that the **kinesthetic-nature** of VR did *not* increase user-engagement in any significant way.
 - Belief of Ability** steadily increases in both groups across the three intervals, and in the post-gameplay survey, though averages are higher in the VG group. After gameplay concluded, the VG group reported a **significantly greater** belief in their ability to win a match: $F(1,30)=.351, p < 0.001$.

DISCUSSION

- It is visible in Figures 1 and Table 2 that when asked questions related to their confidence, subjects in the VG group report a higher average. We believe this has to do with the greater level of exposure and experience subjects in the VG group had with the medium-of-play.
- We found that **differences in duration estimation** (see Table 1) between groups were equally... **inaccurate**.
 - This evidence is supportive of the role of attention in time perception: as cognitive resources are focused on the activity at hand, the passive perception of passing-time becomes less accurate.
- Our findings also support a meta-topic within Psychological research that has been discussed over many happy-hours, and a few publications: **"Why use Virtual Reality technology?"** The resources required for running any experiment using a virtual-reality headset are **costly**. Even for this experiment, which used *Meta Quest 3* headsets, rather than a PCVR set-up, the requirements on researchers were greater than expected. The differences between participants subjected to a video-game console are *not* different enough to justify the costs this experiment required.
 - There is still evidence to support that experiments using Virtual-Reality are still relatively **novel** to the average subject, which may be important depending on the design of the experiment. In this experiment, however, the novelty of VR only resulted in a lower confidence rating, and a higher difficulty rating throughout gameplay..

Table 1. Differences in Duration Estimation

Time Interval	M _{VR}	M _{VG}	t-test	p-value
4 Minutes	7.1 min.	6.2 min.	-0.75	0.456
8 Minutes	9.5 min.	10.8 min.	1.19	0.240
13 Minutes	14.4 min.	16.0 min.	1.01	0.323

Table 2. Differences between Groups, In-Game Survey

Group		VG			VR			Test
Variable		N	Mean	SD	N	Mean	SD	
4-Minute Intake	Belief of Ability	17	3.2	1.3	17	2.7	1.3	$F=1.507$
	Reported Engagement	...	4.1	0.78	...	4.1	0.78	$F=0$
	Reported Difficulty	...	1.6	0.93	...	3.2	1	$F=20.961^{***}$
	Probability of Winning	...	3.9	0.9	...	2.5	1.3	$F=13.166^{***}$
8-Minute Intake	Belief of Ability	...	3.6	0.93	...	2.5	1.1	$F=9.959^{***}$
	Reported Engagement	...	4.2	0.88	...	4.3	0.77	$F=0.171$
	Reported Difficulty	...	1.8	0.73	...	3.2	1.3	$F=16.168^{***}$
	Probability of Winning	...	3.8	1	...	3.2	1.4	$F=1.9$
13-Minute Intake	Belief of Ability	...	3.9	0.7	...	2.8	1.3	$F=10.351^{***}$
	Reported Engagement	...	4.2	0.9	...	4.2	1.1	$F=0.03$
	Reported Difficulty	...	2.9	0.99	...	3.2	1.2	$F=0.615$
	Probability of Winning	...	4.3	0.77	...	3	1.3	$F=12.138^{***}$
After Game	Belief of Ability	...	4.4	0.7	...	3.1	1.3	$F=12.371^{***}$

Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$